Further Integration Techniques

Section 1: Integration by Parts (Repeated & Cyclic)

1 Evaluate $\int x^2 \cos x \, dx$ using integration by parts.

a. $x^2 \sin x + 2x \cos x - 2 \sin x + C$

b. $x^2 \sin x - 2x \cos x + 2 \sin x + C$

c. $-x^2 \sin x + 2x \cos x + C$

d. $\frac{1}{3} x^3 \sin x + C$

2 Evaluate the cyclic integral $I = \int e^x \sin x \, dx$.

a. $\frac{1}{2} e^x (\sin x - \cos x) + C$

b. $\frac{1}{2} e^x (\sin x + \cos x) + C$

c. $e^x (\sin x - \cos x) + C$

d. $-e^x \cos x + C$

3 Evaluate $\int_1^e \ln x \, dx$.

a. 1

b. $e - 1$

c. e

d. 0

Section 2: Trigonometric Substitutions

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4 Which trigonometric substitution is best suited to evaluate $\int \frac{1}{\sqrt{16-x^2}}dx$?

a. $x = 4\sin\theta$

b. $x = 4\tan\theta$

c. $x = 4\sec\theta$

d. $x = 16\sin\theta$

5 Evaluate $\int \frac{1}{x^2+16}dx$.

a. $\frac{1}{4}\arctan\left(\frac{x}{4}\right) + C$

b. $\arctan\left(\frac{x}{4}\right) + C$

c. $\frac{1}{16}\arctan\left(\frac{x}{16}\right) + C$

d. $\frac{1}{4}\ln|x^2+16| + C$

6 Which substitution evaluates $\int \sqrt{x^2-9}\,dx$?

a. $x = 3\sec\theta$

b. $x = 3\sin\theta$

c. $x = 3\tan\theta$

d. $x = 9\sec\theta$

Section 3: Integrals of Powers of Trigonometric Functions

7 Evaluate $\int \sin^3 x\,dx$.

a. $-\cos x + \frac{1}{3}\cos^3 x + C$

b. $\cos x - \frac{1}{3}\cos^3 x + C$

c. $\frac{1}{4}\sin^4 x + C$

d. $-\frac{1}{3}\cos^3 x + C$

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8 Evaluate $\int \cos^2 x dx$.

a. $\frac{1}{2}x + \frac{1}{4}\sin(2x) + C$

b. $\frac{1}{2}x - \frac{1}{4}\sin(2x) + C$

c. $\frac{1}{3}\cos^3 x + C$

d. $\frac{1}{2}x + \sin(2x) + C$

9 Evaluate $\int \tan^2 x dx$.

a. $\tan x - x + C$

b. $\sec^2 x + C$

c. $\tan x + x + C$

d. $\frac{1}{3}\tan^3 x + C$

Section 4: Integration by Substitution (Advanced Forms)

10 Evaluate $\int \frac{x}{\sqrt{x+1}} dx$ using substitution $u = x + 1$.

a. $\frac{2}{3}(x+1)^{3/2} - 2(x+1)^{1/2} + C$

b. $\frac{2}{3}(x+1)^{3/2} + 2(x+1)^{1/2} + C$

c. $(x+1)^{3/2} - (x+1)^{1/2} + C$

d. $\frac{1}{2}(x+1)^2 + C$

11 Evaluate $\int_0^1 \frac{e^x}{1+e^x} dx$.

a. $\ln\left(\frac{1+e}{2}\right)$

b. $\ln(1+e)$

c. $e - 1$

d. $\frac{e}{2}$

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12 Evaluate $\int \frac{1}{x \ln x} dx$ for $x > 1$.

a. $\ln|\ln x| + C$

b. $\frac{1}{2}(\ln x)^2 + C$

c. $\frac{1}{\ln x} + C$

d. $x \ln x - x + C$

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